Due on Tuesday, March 13th.

If pictures are needed/relevant, please provide them with your solutions.

The questions marked **[SELF]** are for yourself and need not be submitted.

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1. Two metallic plates are placed parallel and close to each other. They each have area $A$. They carry opposite charges $+Q$ and $-Q$. Such an arrangement is known as a *capacitor*.

   (a) [**4 pts.**] The plates are large enough that, if we are considering the region between the two plates (far from the plate edges), we can regard each charged plate as an infinite sheet of charge. Show that the magnitude of the electric field in the region between the plates is $E = Q/(A\epsilon_0)$.

   (b) [**3 pts.**] The *capacitance* of a capacitor is the charge on a plate divided by the potential difference between the two plates ($C = Q/V$). If the distance between the plates is $d$, find the capacitance of the arrangement.

2. Maxwell's 2nd equation asserts that the magnetic field has zero divergence.

   (a) [**6 pts.**] The vector fields

   $$\mathbf{V}_1 = C_0 x\hat{\imath} + \frac{C_0}{R}y^2\hat{\jmath} + 3C_0 R\hat{k} \quad \text{and} \quad \mathbf{V}_2 = (C_0R^2)\frac{y\hat{\imath} - x\hat{\jmath}}{x^2 + y^2}$$

   are claimed to describe magnetic fields. (Here $C_0$ and $R$ are positive constants.) Find out whether these claims are compatible with Maxwell's second equation.

   (b) [**SELF**] What are the dimensions of the quantity $R$? (This question means: does $R$ have the units of mass, length, time, charge, or some combination of these?) Explain your answer.

3. An electron (charge$\approx 1.602 \times 10^{-19}$ C; mass$\approx 9.109 \times 10^{-31}$ kg) moving at speed 100 m/s finds itself in a region of constant magnetic field. The magnetic field points perpendicular to the electron velocity and has strength $10^{-4}$ Tesla.

   (a) [**2 pts.**] Find the magnitude of the magnetic force experienced by the electron.

(b) **[2 pts.]** The magnetic force causes cyclotron motion, so that the electron moves around a circular trajectory. Find the radius of the circular trajectory.

(c) **[3 pts.]** How long does it take for the electron to complete one cycle? (i.e., what is the period of the cyclotron motion?)

4. A particle with charge $Q$ and mass $m$ is placed in a region with constant electric field in the $y$-direction, $\mathbf{E} = E\hat{\jmath}$, and constant magnetic field in the $z$ direction, $\mathbf{B} = B\hat{k}$. The particle position coordinates are $x$, $y$, $z$, and its velocity components are denoted as $v_x$, $v_y$, $v_z$.

   (a) **[5 pts.]** Calculate the force $\mathbf{F}$. Write out separately the expressions for its components, $F_x$, $F_y$, $F_z$. (These expressions will involve the quantities $Q$, $B$, $E$, $v_x$, and $v_y$.)

   (b) **[4 pts.]** Using your results for $F_x$, $F_y$ and Newton's 2nd law, $\mathbf{F} = m\frac{d\mathbf{v}}{dt}$, calculate expressions for $\dfrac{dv_x}{dt}$ and $\dfrac{dv_y}{dt}$. You should thus obtain coupled first-order differential equations for $v_x(t)$ and $v_y(t)$.

   (c) **[5 pts.]** If the particle starts at rest, i.e., if the initial conditions are $v_x(0) = v_y(0) = 0$, then the solutions to the above equations are of the form

   $$v_x = \frac{E}{B} - \frac{E}{B}\cos(\omega t) \qquad v_y = \frac{E}{B}\sin(\omega t) \tag{1}$$

   Substituting these into your first differential equation, find an expression for $\omega$. Substituting these into your second differential equation, show that you find the same expression for $\omega$.

   (d) **[5 pts.]** The equations (1) are themselves differential equations for $x(t)$ and $y(t)$, since $v_x = \frac{dx}{dt}$ and $v_y = \frac{dy}{dt}$. Assume that the particle starts at the origin, so that the initial conditions are $x(0) = y(0) = 0$. Find the solutions for $x(t)$ and $y(t)$.

   (e) **[3 pts.]** Sketch $x(t)$ as a function of $t$ and $y(t)$ as a function of $t$. Sketch the trajectiory of the particle in the $x$-$y$ plane.

5. A particle carrying charge $q$ is subjected to uniform electric and magnetic fields, both pointing in the $z$ direction: $\mathbf{E} = E\hat{k}$, $\mathbf{B} = B\hat{k}$.

   At time $t = 0$, the particle is at the origin and has velocity $\mathbf{v} = v\hat{\imath}$.

   (a) **[4 pts.]** At which times does the particle pass through the $z$ axis?

   (b) **[4 pts.]** What are the $z$ values where the trajectory meets the $z$ axis?